CS485 — Data Science and Applications

Spring 2026

Homework 1

Exploratory Analysis · Regression · Probabilistic Modeling

Submission Deadline: April 2nd, 2026

Objective

In this homework, you will apply the concepts covered in Lectures 1–10 to real-world astronomical data from the **Multimodal Universe** platform. You will encode raw data as numerical vectors and matrices, fit regression models using both closed-form and gradient-descent approaches, and reason about uncertainty using probabilistic tools.

Data

Use data from the **Multimodal Universe** platform:
`https://github.com/MultimodalUniverse/MultimodalUniverse`
and the support code in:
`https://github.com/gtsagkatakis/Data-Science-and-Applications-2026`

Select **at least two modalities** (e.g., photometric features and spectral data). Before beginning any analysis, document how your chosen variables are encoded as a data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with n samples and d features, and a target vector $\mathbf{y} \in \mathbb{R}^n$. Explain your encoding choices and any pre-processing steps applied.

Part 1 — Exploratory Data Analysis

Goal: Understand the structure and distribution of your data using the vector/matrix framework.

3.1 Data Encoding & Representation

- Load your dataset using NumPy / Pandas and represent each sample as a row vector. Print the shape of your data matrix $\mathbf{X}$ and describe what each dimension represents.
- Discuss the encoding choices: are features continuous, categorical, or spectral? How did you handle missing values or different scales?

3.2 Similarity & Inner Products

- Compute pairwise **cosine similarities** or **inner products** between a representative subset of samples (e.g. 50–100 objects). Visualise the result as a heatmap.
- Discuss: what does high / low similarity reveal about the objects in your dataset?

3.3 Descriptive Statistics

- Compute mean, median, variance, and standard deviation for each feature.
- Identify features with high variance or skewed distributions.

3.4 Visualisation

- Plot histograms for at least three features. Describe the shape of each distribution (symmetric, skewed, multimodal).
- Plot scatter plots between pairs of features and between features and the target variable $\mathbf{y}$.
- **Discussion:** comment on outliers, variability, and any correlations you observe.

Part 2 — Regression

Goal: Model the relationship between features and a target variable using the optimisation and data-fitting framework from lectures.

4.1 Problem Setup

- Define your input matrix $\mathbf{X}$ and target vector $\mathbf{y}$. Consider the different regression models you can fit (e.g. linear, polynomial, or multivariate).

4.2 Gradient Descent

- Implement **gradient descent** to minimise the Mean Squared Error loss:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{n}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2, \qquad \mathbf{w} \leftarrow \mathbf{w} - \alpha\, \nabla_{\mathbf{w}}\, \mathcal{L}(\mathbf{w})$$

- Plot the **loss curve over iterations**. Experiment with at least two different learning rates α and report your observations.
- **Discussion:** How does the choice of learning rate α affect convergence? What happens when α is too large or too small? At what point do you consider gradient descent to have converged, and how does the loss curve inform this decision?

4.3 Evaluation

- Consider different regression models and report the Mean Squared Error (MSE) and R^2 scores.
- Visualise the data points and the fitted regression line (or surface for the multivariate case).

Part 3 — Probabilistic Modeling

Goal: Characterise the uncertainty in your regression model using the probability and distribution tools.

5.1 Residual Analysis

- Compute residuals:

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$$

- Plot a histogram of the residuals. Describe the shape — is it approximately symmetric? Does it resemble the Gaussian distributions.

5.2 Empirical Gaussian Fit

- Using the sample mean $\bar{r}$ and sample standard deviation s already computed in Part 1, parameterise a Gaussian distribution $\mathcal{N}(\bar{r},\, s^2)$ and overlay its PDF on the residual histogram:

$$f(r) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{(r-\bar{r})^2}{2s^2}\right)$$

- Apply the **68–95–99.7 rule**: verify empirically what fraction of your residuals fall within $\bar{r} \pm s$, $\bar{r} \pm 2s$, and $\bar{r} \pm 3s$. Compare to the theoretical values of 68%, 95%, and 99.7%.

- **Discussion:** How well does the Gaussian fit the residual histogram? What does a poor fit suggest about your regression model?

5.3 Kernel Density Estimation (KDE)

- Apply KDE to the residuals using at least two different bandwidth values.

- **Compare** the histogram, the empirical Gaussian fit, and the KDE on the same plot.

- **Discussion:** What does the KDE reveal that the parametric Gaussian does not? Which gives a better representation of the uncertainty in your model, and why?

Deliverables

Submit a single **Jupyter Notebook** (`.ipynb`) containing:

- All code, clearly commented and organised by Part/Section.

- All plots and numerical results rendered inline.

- Written discussion cells for every **Discussion** item above (use Markdown cells, not code comments).

- A short introduction cell (**Part 0**) documenting your chosen dataset, modalities, and data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$.